

3.5 PROFIT MAXIMISATION

Multiple Choice Questions

1987/AL/I/05

When economists say that a competitive firm faces a perfectly elastic demand curve, they mean

- A. the firm will sell an infinite quantity of output at the going price.
- B. the law of demand is rejected.
- C. all variations in output the firm can make will leave the price unaffected.
- D. the good is not a necessity because the quantity demanded will be zero at a higher price.

1987/AL/I/25

Suppose in a price-taking industry, the market price of a product is \$6. A firm is producing output at the point where the average total cost equals the marginal cost, both of which are \$8. The average variable cost is \$5. In order to maximize wealth, the firm will

- A. keep on producing but reduce its output.
- B. produce zero output.
- C. leave its output unchanged.
- D. increase its selling price.

1992/AL/I/25

The imposition of a lump-sum tax on each firm in a price-takers' market will

- A. decrease the number of firms in the market.
- B. increase the output of each firm that remains in business.
- C. increase the market price of the output.
- D. All of the above.

1995/AL/I/10

Under competitive market equilibrium, all firms will have the same

- A. marginal cost of production.
- B. total cost of production.
- C. profit.
- D. level of output

1998/AL/I/29

The demand curve facing a price-taker is perfectly elastic. This implies that

- A. the market price will not change.
- B. the law of demand cannot be applied in the price-takers' industry.
- C. the market price will not decrease even when a seller increases his output.
- D. All of the above.

2001/AL/I/29

Which of the following comes closest to a product sold in a price-takers' market?

- A. jade
- B. gold
- C. diamond.
- D. pearl

2004/AL/I/12

A firm is said to be a price-taker if

- A. it has the same cost curves as other competing firms.
- B. it cannot alone decide on its output level.
- C. it cannot ask a price higher than the market price and expect to survive.
- D. It receives no profit.

2012/DSE/I/09

Refer to the following table showing the production costs of a price-taking firm.

Total Output (units)	Marginal cost (\$)	Average variable cost (\$)	Fixed cost (\$)
2	4	3	1
3	6	4	1
4	8	5	1
5	10	6	1
6	12	7	1

If the product price is \$6,

- A. the firm will produce 5 units of output to maximize its profit.
- B. the maximum profit the firm can make is \$5.
- C. the average variable cost is \$5 when the firm maximizes its profit.
- D. the minimum total cost of the firm is \$13.

2014/DSE/I/06

The following table shows the cost-output relationship of a price taker.

Output (units)	Total cost (\$)
1	20
2	45
3	75
4	110
5	150

Given that the profit maximizing output of the firm is 3 units. If the fixed cost increases by \$5, the price of the good and the output will be _____ and _____ respectively.

- A. \$ 25 2 units
- B. \$ 30 2 units
- C. \$ 30 3 units
- D. \$ 75 3 units

2015/DSE/I/05

The schedule below shows the marginal cost of a firm in a perfectly competitive market.

Output (units)	3	4	5	6	7	8	9
Marginal cost (\$)	5	5.5	6	6.5	7	8	9.5

Suppose the rental of the machine hired, a fixed cost, increased by \$2 while the marginal cost for each unit of the output is reduced by \$0.5. As a result, the profit-maximising output of the firm _____ while the profit of the firm _____.

- A. increases decreases
- B. increases may increase or decrease
- C. decreases decreases
- D. decreases may increase or decrease

2016/DSE/I/12

The following table shows the cost-output relationship of a price-taker.

Output (units)	Average cost (\$)
2	20
3	21
4	22
5	23
6	24
7	25

Suppose the market price is \$25. What are the profit-maximising output and profit respectively?

- A. 4 units and \$0
- B. 4 units and \$12
- C. 7 units and \$0
- D. 7 units and \$12

2017/DSE/I/08

The following table shows the cost-output relationship of a price-taker.

Output (units)	Average cost (\$)
5	10
6	11
7	12
8	13
9	14

The market price is \$20. Suppose the average cost has increased by \$2 for every unit of output. The profit of the firm will

- A. increase by \$16
- B. increase by \$14
- C. decrease by \$16
- D. decrease by \$14

2018/DSE/I/06

The following table shows the cost-output relationship of a profit-maximising firm. The firm has a fixed cost of \$5.

Output (units)	1	2	3	4	5	6
Average cost (\$)	7	6	7	8	9.6	12

Which of the following statements is correct?

- A. The optimal scale of production is 2 units.
- B. There exist diseconomies of scale after the third unit of output is produced.
- C. If the market price is \$8, the output is 4 units.
- D. If the market price is \$9, the profit is \$6.

2019/DSE/I/10

The following table shows the cost-output relationship of a price-taking firm.

Total output (units)	Average cost (\$)
3	10
4	15
5	20
6	25
7	30

If the profit-maximising output is 6 units, the market price will be _____.

- A. \$5
- B. \$25
- C. \$30
- D. \$50

2020/DSE/I/9

Refer to the following table about the cost-output relationship of a profit-maximising firm.

Quantity (units)	1	2	3	4	5
Average cost (\$)	5	6	7	8	9

If the average cost of every unit of output decreases by \$1 and the market price is \$8,

- A. the minimum cost of production is \$18.
- B. the profit maximising output is 3 units.
- C. the maximum profit is \$10.
- D. the marginal cost of every unit of output will remain unchanged.

2021/DSE/I/10

Because of the outbreak of COVID-19, the government of a country announced a temporary ban of dine-in services in restaurants. Which of the following statements about restaurants under the temporary ban is correct?

- A. If the expected sales revenue generated from takeaway food exceeds the variable cost of running a restaurant, the restaurant will continue to operate.
- B. With the extremely high cost of breaking the rental contract, the fixed rental will affect whether the owners continue to operate their restaurant.
- C. A decrease in prices of takeaway food will necessarily decrease the sales revenue of the restaurants.
- D. Unpopular restaurants will close down before popular ones.

2021/DSE/I/12

The following table shows the cost-output relationship of a price-taking firm.

Quantity (units)	Average cost (\$)
1	5
2	4
3	4
4	4.5
5	6

Based on the above table, when the market price falls from \$6 to \$4, we can conclude that

- A. the firm starts to enjoy economies of scale.
- B. the total cost of the firm will decrease from \$30 to \$12.
- C. the profit of the firm will decrease by \$6.
- D. the quantity supplied will decrease from 5 units to 3 units.

2019DSE/I/04

If the fixed cost of a price-taking firm decreases by \$200 and the marginal cost of each unit of output increases by \$10, the profit-maximising output will _____ and the profit _____.

- A. increase will decrease
- B. increase may increase or decrease
- C. decrease.....will increase
- D. decrease.....may increase or decrease

2022/DSE/I/11

Winnie operates a café in Sham Shui Po. To maximise her profit, Winnie will produce more output if

- A. water charges for commercial use fall.
- B. business registration fee falls.
- C. the monthly rental payment of the café rises.
- D. the wage rate of part-time waiters increases.

2023/DSE/I/09

The following table shows the data of a price-taking garment factory. The fixed cost of the factory is \$ 10.

Output (units)	Total cost (\$)
6	48
7	63
8	80
9	99
10	120
11	143

Suppose the market price of garments rises from \$17 to \$19, and the fixed cost of the factory rises by \$2. As a result, the profit-maximising output of the factory would _____ while its profit would _____.

- A. remain unchanged.....increase to \$56
- B. remain unchanged.....increase to \$70
- C. increase by 1 unit.....increase to \$70
- D. increase by 1 unit.....increase to \$82

2024/DSE/I/10

The table below shows the production costs of a price-taking firm.

Total output (units)	Average cost (\$)
1	20
2	25
3	30
4	35

Suppose the fixed cost of the firm is \$5 and the product price is \$30. The firm's maximum producer surplus is _____.

- A. \$0
- B. \$5
- C. \$10
- D. \$15

2025/DSE/I/07

The table below shows the cost-output relationship of a firm in a perfectly competitive market. The fixed cost of the firm is \$6 and the market price is \$20.

Output (units)	1	2	3	4	5	6	7
Total variable cost (\$)	4	12	24	40	60	84	112

Suppose the fixed cost decreases by \$4 and the marginal cost for each unit of output increases by \$4. As a result, the firm's profits will _____.

- A. remain unchanged
- B. decrease by \$6
- C. decrease by \$ 12
- D. decrease by \$ 16

Short & Structured Questions

SP/DSE/II/9(c)

Assume the market of electronic watches is perfectly competitive.

Suppose there is an improvement in technology that lowers the marginal cost of producing electronic watches. The schedule below shows the marginal costs of a firm which produces electronic watches before and after the technological improvement.

Output (Units)	Marginal cost (\$) <i>before</i> the technological improvement	Marginal cost (\$) <i>after</i> the technological improvement
500	70	60
600	80	70
700	90	80
800	10	90
900	110	10

- (i) Is marginal cost a fixed cost or a variable cost? Explain. (2 marks)
- (ii) Suppose the market price of electronic watches was \$80 before the technological improvement. What is the output of the firm at this market price? (1 mark)
- (iii) Suppose the market price of electronic watches fell to \$70 after the technological improvement. What is the output of the firm at this market price? (1 mark)

2024/DSE/II/11(b)

Ann is the owner of a coffee shop in Hong Kong.

- (b) The landlord of Ann's coffee shop premises raised the monthly rental from \$70 000 to \$85 000. Ann said, 'Despite the increase in the monthly rental, the quantity of daily output of my coffee shop remains unchanged.'
Explain Ann's output decision with reference to the profit-maximising condition under perfect competition. (3 marks)